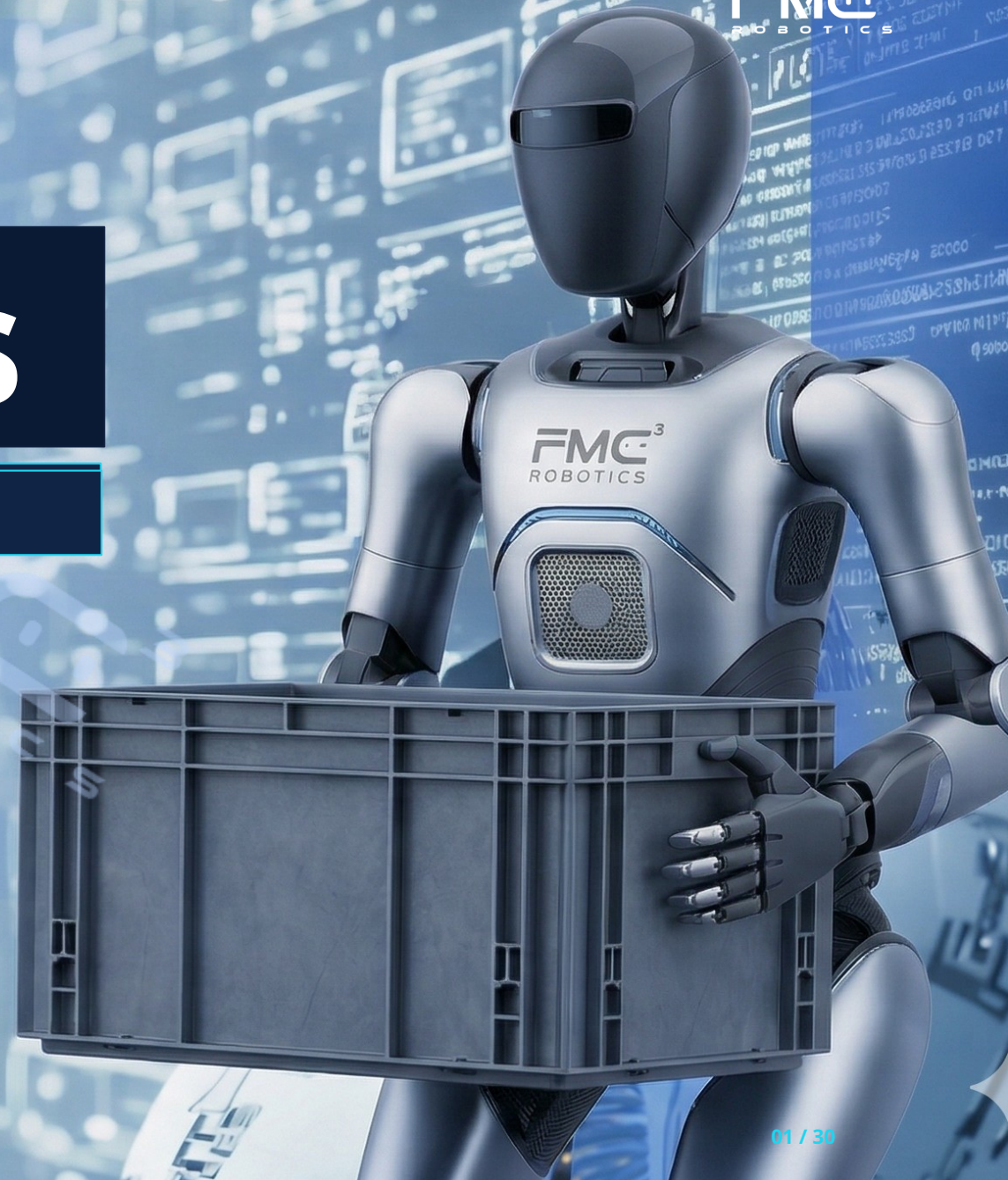


FMC³ Robotics

The Embodied AI Brain from Europe

Product Manual
Mar, 2026
Confidential



Positioning

- Maximizing the commercial value of Embodied AI in developed Western markets by leveraging German commercial scenarios and Chinese supply chain excellence.
- Establishing crossscenario standards for Embodied AI through a "Physical AI + ForceTouch Closed Loop" to achieve fullstack autonomy.



Eur

Schloßlande 26, Ingolstadt
Bavaria, Germany

Strategic HQ: Direct market access with a focus on realworld deployment.

Data Moat: Compliance with EU AI Acts and GDPR data regulations.

Training & Delivery: Building the data training closed loop.



China

Jia Xing & Shanghai
China

R&D & Design: Development of highfidelity data collection systems.

Hardware Supply Chain: Comprehensive upstream and downstream robotics resources.

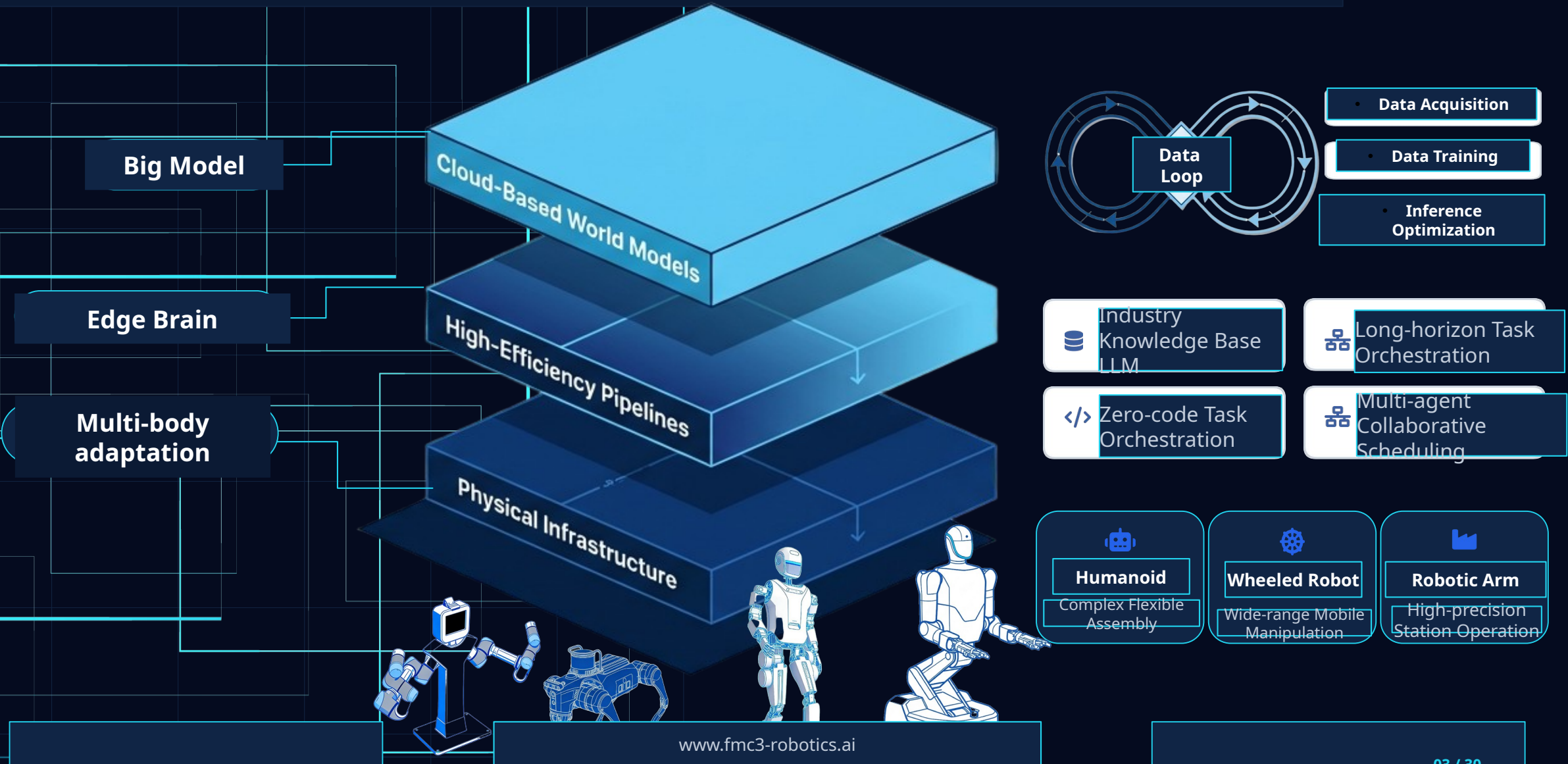
Pilot & Testing: Rapid POC/MVP iterations and closedloop training tests.

HQ

Product
&Engineering

China office
RD&Product
design

One Brain filt all



Core team

A Team of European Intelligent Systems Scientists



Dr. Wang Cheng

CEO

- Dr.-Ing. from Technical University of Munich (TUM).
- Expert in Autonomous Driving.
- Architect of Audi's Intelligent Driving System.
- Proven track record in the mass production and delivery of over 1 million intelligent driving vehicles.



Prof. Dr. Giesler Bjoern

CTO

- PhD in Computer Science from KIT; a leading authority in AI architecture.
- Holds 50+ core patents in sensing and data training.
- Extensive experience managing international R&D teams of 500+ members.



Fan David, PhD

COO

- Former President of Marelli China.
- Managed business operations with annual revenues exceeding \$1.5 billion.
- Led multiple post-M&A organizational integrations and supply chain restructurings.



Dr. Huang Dong

CPO

- Dr.-Ing. in Computer Science from KIT.
- Former Scientist at Siemens Labs and CTO of TerraIT.
- 15+ years of expertise in data simulation and Sim-to-Real domains.



Obermeier Florian

MD Germany

- Co-founder of SANEON GmbH.
- 16+ years of B2B experience with OEMs and Tier-1 automotive suppliers.
- Global track record serving clients across Europe, Asia, and the US.



Localized Data Closed-Loop Architecture (Germany/Europe)

Local Data Loop



Local Collection

Zero Cross-border Transfer. All industrial data stored 100% within German sovereign data centers.

Encrypted



Privacy Computing

Auto-Anonymization. Real-time de-identification of facial & environmental metadata; GDPR compliant.

Anonymized



Federated Learning

Gradient-only Sync. Local data remains on-premise; only model parameters are transmitted.

Secure ML



Compliant with the EU AI Act

Fully compliant with regulatory requirements for high-risk AI systems, featuring transparent and interpretable auditing capabilities.



Principle of data minimization



right of user to be forgotten



storage limitation principle



Limitation Principle of Purpose

Entering the European Industry: Building a Moat Based on "Deterministic, Safe, and Compliant" Standards



Collaborative Safety & Compliance

Machinery Regulation (EU)
Core Operational Safety

EMC Directive
Device Interference-free
Operation

RED: Wireless Communication
Security

EU AI Act
AI Algorithm Governance

The Four Hard Requirements for Embodied AI Robots under the "Machinery Regulation"

1



1. Safety of Control Systems

- Resilience against operational stress and interference.
- Must not lead to hazardous situations

2



2. Special Requirements for Autonomous Mobility

- Sensing & Stopping: Automatic stop upon detection of humans.
- Motion Stability: Anti-tip and anti-loss-of-control measures.

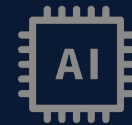
3



3. Risk Monitoring & Cybersecurity

- Defense against malicious network manipulation.

4



4. Software Safety & Self-learning Learning Boundaries:

Implementation of strict learning constraints.
OTA Assessment: Mandatory safety re-evaluation for all remote software updates.

Our Partners and Clients

Technical
University
of Munich



NMI
Nationalwissenschaftliches
und Ingenieurwesen Institut



KIT
Karlsruhe Institute of Technology

FAU

Friedrich-Alexander-Universität
Erlangen-Nürnberg



KAUTEX

A Textron Company



Technische Hochschule
Ingolstadt

INOVANCE



FOURIER



HIKAUTO

AGIBOT

Products

FMC³ Brain+ODM

Robo dog



Commercial & Industrial Cleaning



Workbot



Cobot



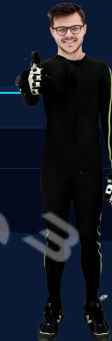
GreetingBot



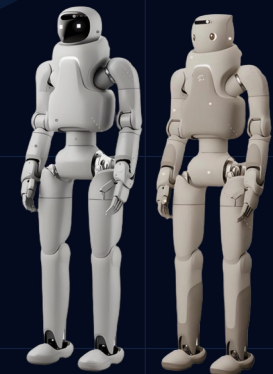
Industrial Service Robot



FMC³ Brain + FMC³ Data Acquisition Suit



Humanoid Home Service Robot



* All hardware, software, solutions, and related services can be provided either as standalone component or as fully integrated turnkey packages, depending on customer requirement.

Products

FMC³ Brain+Data Acquisition Suit

FMC³ Brain



Main Computing Chips:

Nvidia Orin / Thor

Computing Power:

275 TOPS (Scalable)

Interface Standards:

EtherCAT / CAN FD

Power Consumption:

15W 60W (Dynamic)



Realtime Teleoperation & Emergency Takeover

- LowLatency Streaming: Supports crossborder remote operations.
- Dataintheloop: Realtime cleaning and ingestion of teleoperation data
- Safety Critical, industrialgrade safety takeover mechanism.



FullLink Data Toolchain

- Workflow: Collection → Autolabeling → Training Closed Loop.
- Compatibility : Fully compatible with NVIDIA Omniverse and Isaac Sim.
- Heterogeneous Data Fusion: Seamless integration of Realworld + Simulation data.

Core Data Set

Industrial Haptic MultiAgent Dataset

- Proprietary industrialgrade dataset featuring haptic/force modalities.
- Covers 3,000+ core industrial processes, including precision assembly, flexible sorting, and surface polishing.



Joint R&D: Shanghai Jiao Tong University (SJTU) Robotics Institute

Products

FMC³ Brain+Data Acquisition Suit

FMC³ Omni-Grab : Ultra-Fidelity, Body-Free Human-Ego-Centric Data Capture System

Active IR Stereo RGB-D Headset



Delivers first-person 3D depth spatial perception. Both depth and color run at 1280×720 @ 30fps with an integrated 6-axis IMU (50Hz) for dynamic tracking.

First-person geometric and semantic depth alignment.

400-Point High-Density Haptic Glove



32 sensors per finger, trigger force as low as 30g, max range 1500g, with 200Hz haptic stream high-speed export.

Delivers the force-feedback ground truth most critical for imitation learning

Portable Data Sync Hub



Supports 16 hours of continuous single-session recording

Handles global millisecond-grade hardware time alignment, local caching, and data security

DATA REFINERY PIPELINE

End-to-End Automated Processing Pipeline

STEP 1

High-Precision Synchronized Capture:

Raw video, IMU, haptic, and EMG streams are hardware-stamped and losslessly aligned to disk.

STEP 2

Automatic Scene Segmentation & Slicing:

ORB-SLAM3-powered segmentation automatically slices semantically rich task episodes by action and odometry.

STEP 3

Smart Cleaning & Multi-Dimensional QA:

Physics-plausibility validation automatically filters corrupted frames, body penetrations, and violent jitter.

STEP 4

Auto-Annotation & Programmatic Augmentation:

Integrates YOLO and SAM2 for object detection, plus HaMeR for hand pose and contact torque estimation. Supports lighting and viewpoint augmentation.

STEP 5

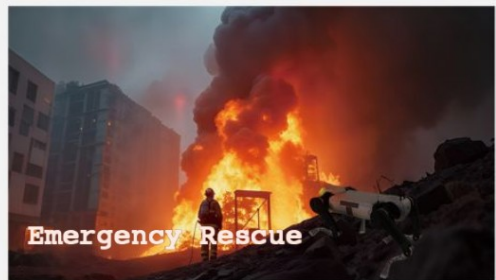
Standardized Packaging & Universal Export:

One-click export to ROS Bag, PyTorch Dataset, and TFRecord- ready for mixed-model training out of the box.

Products

FMC³ Brain+ODM

Robo dogD1
MAX PRO



Highlights

- **Ultrahigh Payload**

Supports sustained effective payload of **50kg**, meeting the demands of heavyload scenarios such as integration with large payload modules, overweight transportation, and material handling.

- **Reliable Battery Life**

Equipped with a **2160Wh highcapacity battery** featuring exceptional power conversion efficiency. It delivers **2.5 hours of continuous operation under rated load**, and up to **5.5 hours (industryleading) in idle conditions**. The system also supports **hotswappable batteries**, with an optional autonomous charging dock available.

- **Outstanding Mobility Performance**

Boasts exceptional agility and allterrain adaptability. Capable of smoothly climbing **30cm continuous staircases and open industrial stairs**, with a robust selfrighting mechanism in place.

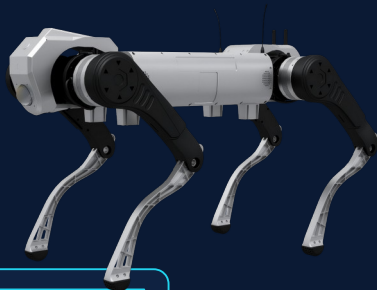
Superior Environmental Adaptability

With **IP67 industrialgrade protection**, it operates reliably across a wide temperature range of **20°C to 55°C** and in water depths up to **20cm**. Built to withstand extreme cold, scorching heat, wind, and rain, ensuring stable performance in harsh conditions.

Products

FMC³ Brain+ODM

Robo dogD1
MAX PRO



Parameters

Standing Dimensions (L × W × H): 1230mm × 530mm × 730mm
 Prone Dimensions (L × W × H): 1313mm × 530mm × 300mm
 Total Weight (Including Battery): 68kg
 Operating Temperature: 20°C ~ 55°C
 Protection Class: IP67
 Battery: Rated Capacity 45Ah, Voltage 48V
 Charging Time: 4.5h (2h Fast Charging Optional)
 Battery Runtime: 5.5h (No Load), 2.5h (Full Load)
 Maximum Range (Full Load): 13.5km

Sensing Unit

LiDAR: 1 × 96channel LiDAR (Frontmounted), Coverage Angle: 360° × 90°, Coverage Diameter: 120m, Accuracy: 1.5cm
 WideAngle Camera: 1 × WideAngle Camera at the Front and 1 at the Rear, Industrial Grade 2Megapixel, Field of View: DFOV 150°, HFOV 130°, VFOV 73°
 IMU: Standard Configuration (SC)

Functions

- Standard Motions: Standing, Lying Down, Damping, Force Unloading; Stair Climbing (Up/Down), Obstacle Crossing
- RealTime Image Transmission: Supported
- Secondary Development: Supported
- 4G Communication: Optional

Performance Parameters

Continuous Walking Speed: 1.5m/s
 Maximum Speed: 3m/s
 Rated Payload: 50kg
 Continuous Stair Climbing Height: 30cm
 Maximum Climbing Slope: 45°

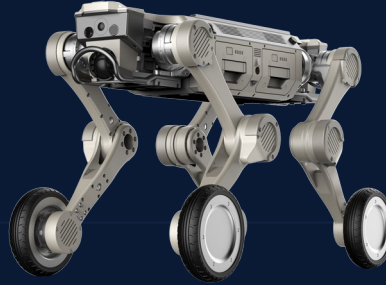
Robot Joint Parameters

Maximum Joint Torque: 180N·m
 Joint Motion Range (Body): 45° ~ 45°
 Joint Motion Range (Thigh): 200° ~ 20°
 Joint Motion Range (Calf): 30° ~ 160°

Products

FMC³ Brain+ODM

Robo dogD1 MAX



Highlights

1. Core Performance Advantages

- Engineered for industrialgrade performance
- Low selfweight paired with high payload capacity
- Longlasting battery life for extended operations
- Robust environmental protection design
- Agile yet stable locomotion for demanding tasks

2. Intelligent Locomotion & Terrain Adaptability

- Reinforcement learningbased motion control model
- Builtin capabilities: impact resistance, disturbance rejection
- Fall selfrecovery mechanism for reliable field use
- Dynamic adaptive control for complex environments
- Autonomous adaptation to diverse terrains and road conditions

3. FullScene Perception & Navigation

- Multimodal perception system: vision, LiDAR, ultrasonic, and INS
- Environmental detection and mapping capabilities
- Autonomous navigation in unstructured spaces
- Realtime target recognition and tracking
- Enables intelligent, collisionfree operations

4. MultiScenario Application Capabilities

- Rich set of power and communication interfaces
- Versatile for industrial and public safety use cases:
- Security patrol & surveillance
- Power grid inspection
- Logistics and heavy payload transport
- Onsite task execution & response
- Personnel search & rescue (SAR)

Products

FMC³ Brain+ODM

Robo dogD1 MAX

Basic Information

Material: Aluminum Alloy + HighStrength Engineering Plastic

Standing Dimensions (L × W × H): 920mm × 480mm × 570mm

Prone Dimensions (L × W × H): 930mm × 630mm × 200mm

Total Weight (Including Battery): 40kg

Operating Temperature: 20°C ~ 55°C

Protection Class: IP67

Battery: Rated Capacity 10Ah, Voltage 54V

Charging Time: < 2h

Battery Runtime: 5h (No Load), 2.5h (Full Load)

Maximum Range: 29km (No Load), 18km (Full Load)

Performance Parameters

Maximum Speed: 8m/s

Rated Payload: 25 ~ 30kg

Continuous Stair Climbing Height: 22cm

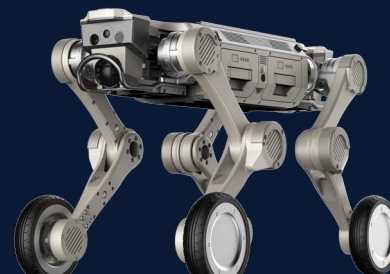
Maximum Climbing Slope: 45°

Jumping Height Off the Ground: 80cm

Maximum Obstacle & Ditch Crossing Height: 80cm

Loaded Obstacle Crossing Height: 50cm

Minimum Passable Width: 50cm



Robot Joint Parameters

Maximum Joint Torque: 150N·m

Joint Motion Range (Body): 30° ~ 40°

Joint Motion Range (Thigh): 160° ~ 160°

Joint Motion Range (Calf): 160° ~ 160°

Sensing Unit

LiDAR: 1 × 192channel LiDAR at the front and 1 at the rear, with a coverage angle of 360° × 90° and a coverage diameter of 120m

Camera FOV: 1 × FPV camera at the front and 1 at the rear, industrialgrade 8megapixel, field of view: DFOV 122°, HFOV 111°, VFOV 70°

Fill Light: 4 × Fill Lights, 1 on each side of the FPV cameras

IMU: 1 × AutomotiveGrade IMU

RTK Module: 1 × GNSS RTK

Ultrasonic Sensor: 1 × Ultrasonic Sensor on each side, supporting obstacle recognition within 5m

Products

FMC³ Brain+ODM

Designed for specialized and industrial applications

Industrial Grade Compact Quadruped Robot Designed for specialized and industrial applications requiring reliability, mobility, and flexible deployment.

Robo dog D1 Ultra



Highlights

Tackle various complex terrains

- Reinforcement learning (RL) based gait control enables autonomous adaptation to diverse and complex terrains, with built-in self-balancing, anti-tipping, and disturbance rejection capabilities
- Maximum running speed of 3.7 m/s, maximum climbing/descending slope angle $\geq 30^\circ$
- Forward or vertical jump liftoff height up to 35 cm
- Supports continuous climbing of stairs with a maximum riser height of 16 cm

Robot Joint Parameters

Maximum Joint Torque: 48N·m

Joint Motion Range (Body): $28^\circ \sim 28^\circ$

Joint Motion Range (Thigh): $170^\circ \sim 66^\circ$

Joint Motion Range (Calf): $35^\circ \sim 156^\circ$

Sensors

IMU: Standard Configuration (SC)

Camera FOV: 1 Camera with 1920×1080p Resolution, Field of View: DFOV 122° , HFOV 111° , VFOV 70°

Performance Parameters

Maximum Speed: 3.7m/s

Rated Payload: 5kg

Continuous Stair Climbing Height: 16cm

Maximum Climbing Slope: 30° in Standard Environment, Up to 40° in Extreme Conditions

Jumping Height Off the Ground: 35cm

Products

FMC³ Brain+ODM

Designed for specialized and industrial applications

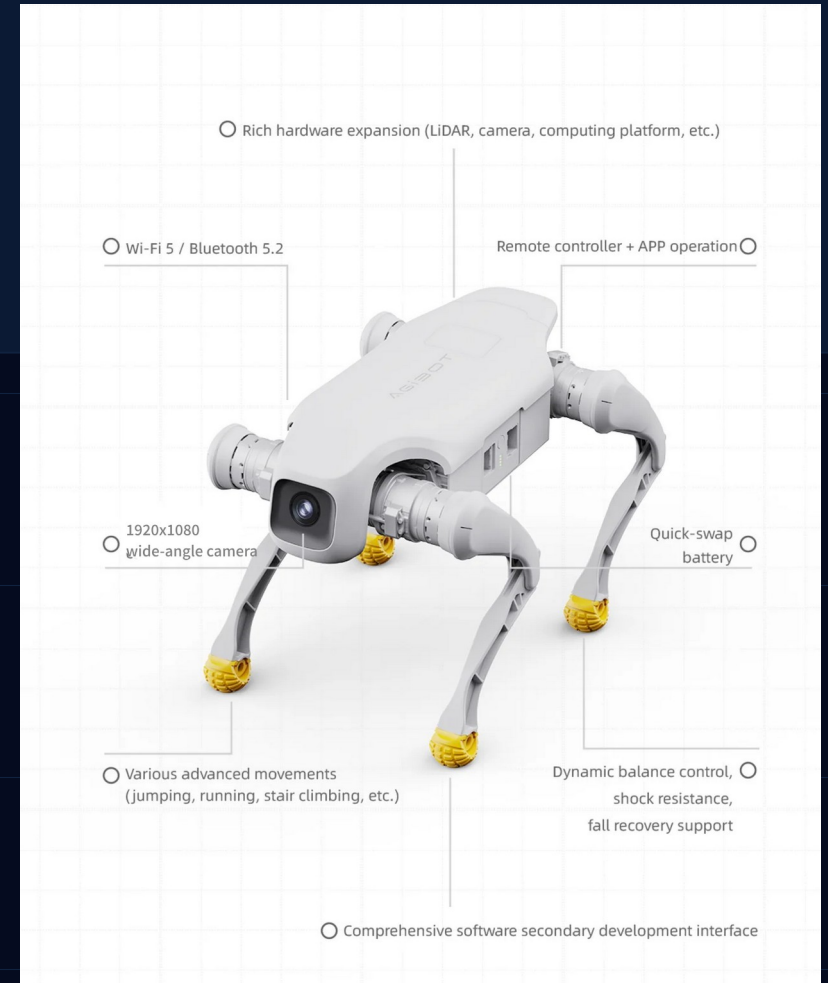
Industrial Grade Compact Quadruped Robot Designed for specialized and industrial applications requiring reliability, mobility, and flexible deployment.

Highlights

- **Powerful Locomotion** to Tackle All Complex Terrains
- Reinforcement learning (RL)-powered gait control enables autonomous adaptation to diverse and challenging terrains.
- Built-in capabilities: self-balancing, anti-tipping, and disturbance rejection for stable performance in dynamic environments.

- Prone Dimensions (L × W × H): 675mm × 435mm × 145mm
- Total Weight (Including Battery): 15.5kg
- Operating Temperature: 0°C ~ 40°C
- Protection Class: IP54
- Battery: Rated Capacity 4.6Ah, Voltage 43.2V
- Charging Time: 1h
- Battery Life: 1-2h
- Maximum Range: 6km

Robo dog D1 Pro



Products

FMC³ Brain+ODM

Robo dog



Maximum Joint
Torque: Approx.
180 N·m



Extreme Locomotion Capability:
Maximum Step Height: **30 cm**
Maximum Slope Angle: **45°**



Comprehensive Payload Capacity:
Continuous Walking Payload:
25 kg
Maximum
Standing Payload:
100 kg



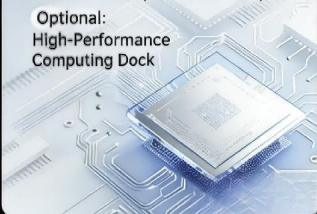
Material:
Aluminum Alloy + High-Strength
Engineering Plastic



Maximum Running
Speed:
5 m/s



Computing Module:
Standard:
8-Core High-Performance CPU
(Platform Functions)
+ Intel Core i7 (for User Development)
Optional:
High-Performance
Computing Dock



Functions

- SLAM Localization and Mapping
- Autonomous Navigation
- Omnidirectional Obstacle Avoidance
- Point Cloud Surround View
- Intelligent Management and Scheduling System
- Autonomous Charging
- High reliability
- Aftersales support

Parameters

Standing Dimensions (mm)

820 × 440 × 570

Weight	37 kg
Continuous Working Payload	25 kg
Max. Continuous Step Height	30 cm
Max. Step Height	80 cm
Limit Test Speed	5 m/s
Max. Operating Speed	3.7 m/s
Max. Climbing Angle	Approx. 45°
Trench Crossing Width	
NoLoad Endurance Time / Range	5 h / 20 km
Effective Payload Endurance Time / Range	3 h / 12.5 km

Battery Capacity

907 Wh

Products

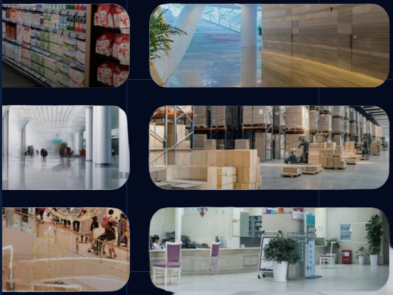
FMC³ Brain+ODM

Commercial &
Industrial Cleaning



Application Scenarios

- Large supermarkets
- Office buildings
- Transportation hubs
- Factories
- Commercial
- Hospitals



Functions

- Allinone vacuuming & scrubbing
- Laservision fusion positioning
- Auto charge, water refill/drain, selfcleaning
- Highefficiency edgetoedge cleaning
- Low maintenance, long service life

Parameters

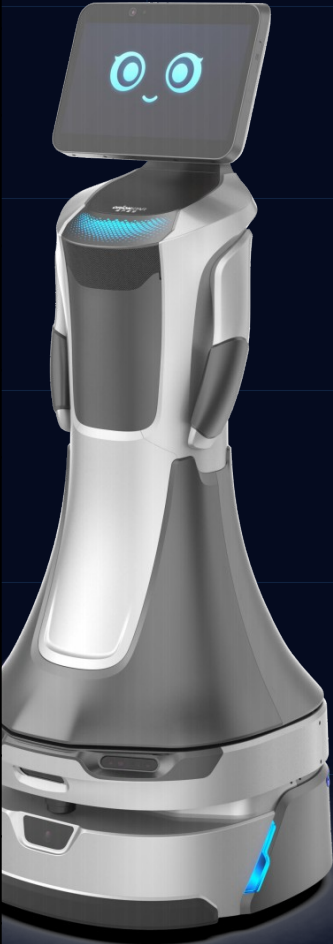
Garbage collection rate	95%	Overall noise	<68 db(A)
Machine weight	170 kg	Extra large volume of water tank	90 L
Selfcleaning dirty water tank	4 min	Floor scrubbing endurance	3 h
Battery specification	24V 80 Ah	Scrubbing pressure	25 kg
Maximum efficiency	1980 m ² /h	Scrubbing width	550 mm
Lifespan of Fan	10000 h		
Water tank capacity	8 L		
Water filling speed		710 L/min	
Dirty water discharging speed		1015 L/min	
Maximum power		1800 W	
Rated input voltage		200240 V	

Applicable Floor Types



GreetingBot Star Key Specifications

Size	558mm×525mm×1350mm
Net Weight	56kg
Screen Size	14-inch; 1080 HD
Chest Part Load Weight	5Kg
Narrowest Passage Width	70cm
Hardware Platform	Qualcomm 8-core chip + Industrial-grade MCU
Navigation Sensor	Li DAR*1 + Depth Vision Camera*3 + Fisheye Camera*2 + Infrared Camera*2 + Odometer + IMU*1
Microphone Array	6-Mic arrays, 360° sound source positioning, 5 m sound range
Battery Life	12 H (typical cruising scene)
Charging Time	4.5 H
Supports Secondary Interface	USB 2.0 A port*2, Type-C port*1; LAN 100Mbps Ethernet port *1; DC power port 12V 3A



Reception Service

Reception of Visitors, Proactively Greeting

Tour Guide

Intelligent Guiding, Obstacle Avoidance

Dynamic Promotion

Dynamic Broadcasting, Attract Customers

Intelligent Q&A

Personalized Scenario Chatbot Service



Products

FMC³ Brain+ODM

Workbot



Application Scenarios

- Automotive: LineSide Feeding, Flexible
- CNC: Machine Tending, MultiMachine
- 3C: Component Handling, MultiSKU
- Logistics: Tote/Pallet Handling, Automation
- General Industry: Flexible Line Retrofit

Functions

- Unified Integrated Control, No Interface Boundaries
- SelfDeveloped Positioning Compensation, EnvironmentResilient
- Intuitive HMI, OneClick Command Execution
- Dual LiDAR + Dual Vision + Collision Detection, Enhanced Safety
- Rapid Battery Swap, 24/7 Continuous Operation
- Modular Chassis + Backpack Design, Maximized Flexibility

Parameters

Dimensions (L×W×H)

950×630×850 mm (excluding arm)

Manipulator

Reach 1327 mm, payload 12 kg (incl. end effector)

Weight & Payload

Weight < 300 kg, usable payload 150 kg

Speed

Arm TCP ≤1.8 m/s, travel ≤1.2 m/s

Mobility

Obstacle ≤10 mm, gap ≤20 mm

Passage Width

Travel ≥850 mm, turning ≥1250 mm

Protection

Body splashproof, arm IP68

Accuracy

Station ±10 mm, comprehensive ±0.5 mm

Endurance

Single battery 3 h, dual batteries 6 h

Cycle Time

20 s

Radar Layout

Diagonal dual radar

Products

FMC³ Brain+ODM

Seven-degree-of-freedom bionic arm

- Dual-arm rated payload 7Kg, maximum 10Kg
- Seven-axis anthropomorphic configuration, low weight-to-payload ratio
- Seven torque sensors for compliant force control

Embodied Intelligence

- Supports high computing power of 2070 Tops
- Compatible with mainstream Embodied Large Models
- Compatible with multiple data acquisition devices

Quick-change End Effector

- Supports quick-change for dexterous hands, grippers, and fixtures

Navigation system

- Multi-sensor fusion navigation: 3D LiDAR + 2D LiDAR + 3D camera

Omnidirectional Mobile Chassis

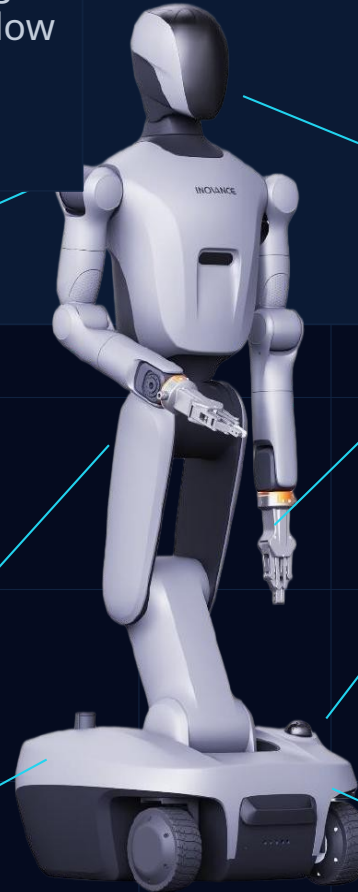
- Compact dimensions: 594X591mm
- Operating in narrow workstations and aisles
- Omnidirectional movement, in-place rotation
- Maximum Speed: 1.5m/s

Full-body Motion Control System

- Full-body Coordinated High-frequency Motion Control
- Supports ECAT hard master station, high reliability
- Waist pitch and 180° rotation, for larger and more efficient working space

Power management system

- Endurance time: 8h
- Supports hot-swap battery exchange



Application Scenario



Production Line Loading and Unloading



Logistics Sorting



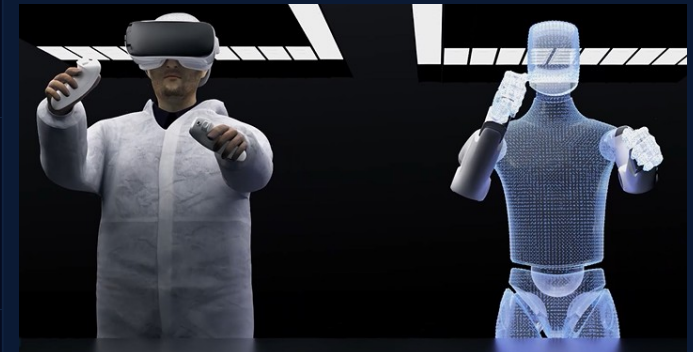
Unmanned Retail



Wellness and Elderly Care Collaboration



Research and Education



Data Collection Training

Humanoid upper limb, compliant control, safe human-robot collaboration



Rated/Max. load: 3.5 kg/3.5 kg
Weight: 7.5 kg

- Humanoid configuration for confined space operation, reducing wrist space requirements by 50% +
- Smooth motion control, stable grasping, maximum overshoot $\leq 1\text{mm}$, settling time $\leq 0.5\text{s}$
- Seven-joint force sensing, high-precision force control, 1000Hz full-body force control frequency



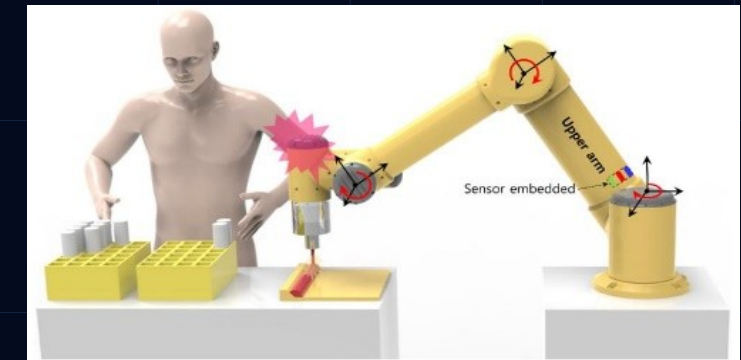
- Vibration suppression: High-efficiency vibration damping, precise start-stop, precise compensation of joint flexibility, reduced vibration, no jitter during heavy-load operations



- Force-controlled insertion: Active force-position coupling for correction, dynamic adjustment of sub-millimeter deviations, ensuring successful completion of precision insertion tasks

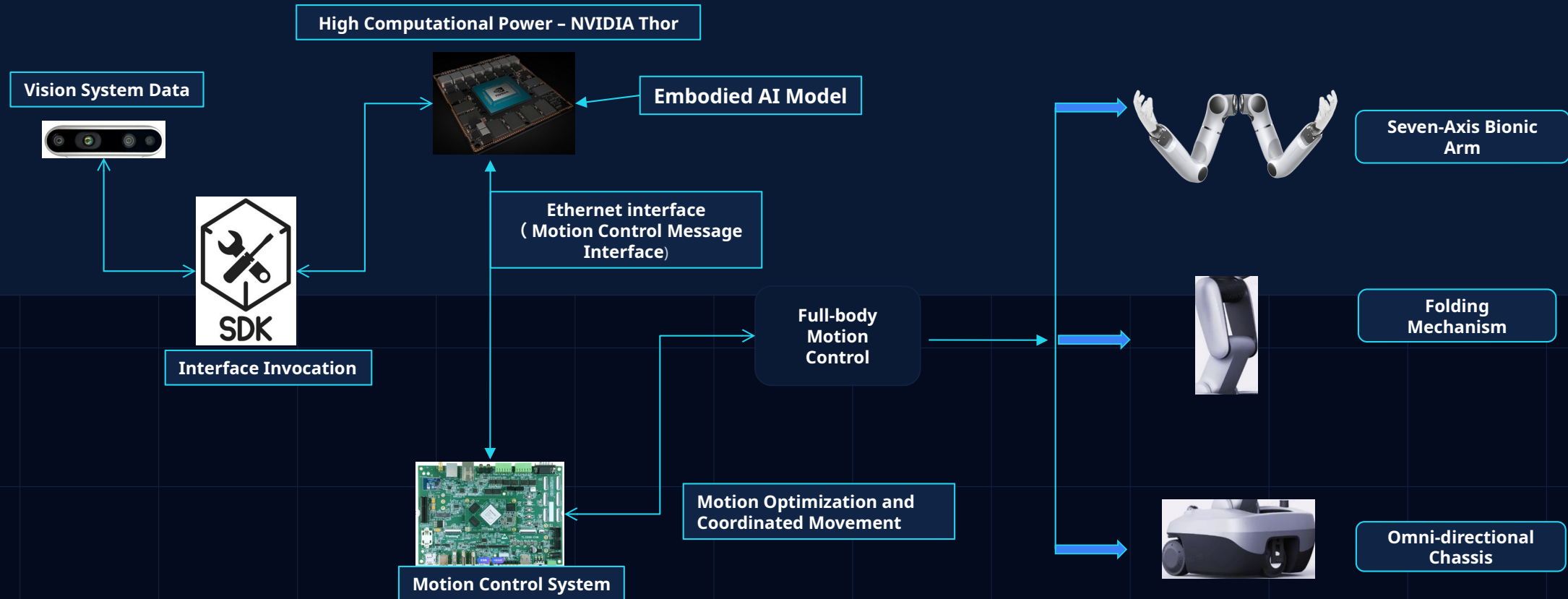


- Suitable for operation within a 60 x 60 cm workspace, with minimal motion interference, compact wrist design for greater reachable positions



- Safe collaboration: $< 1\text{ N}$ high-precision detection, stop within $< 1\text{s}$ in case of collision, ensuring human safety

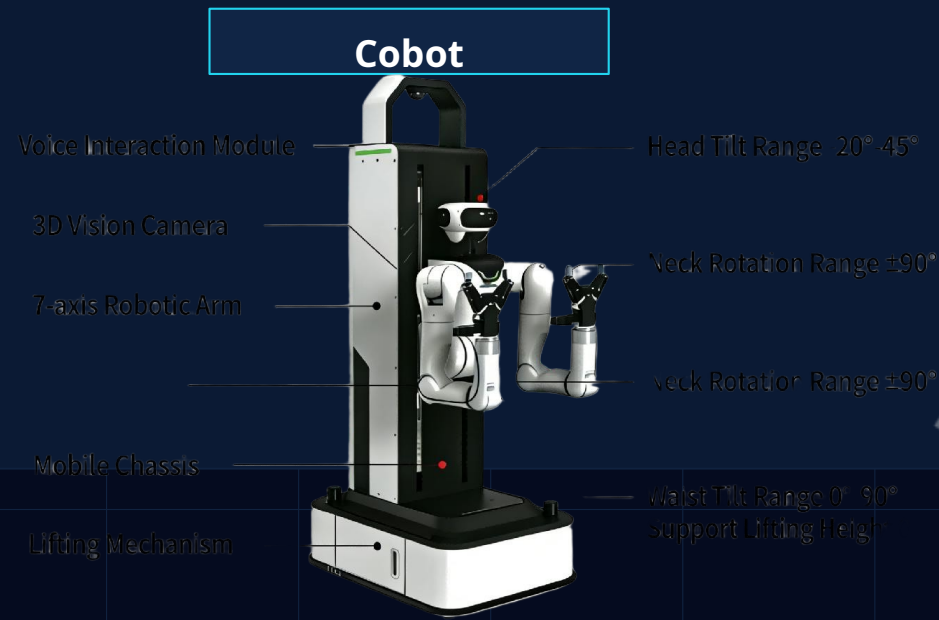
Refined and User-Friendly Infrastructure – Customer focus is on AI model deployment and research



- AI Hardware Platform with Underlying Motion Control Hardware Platform, Operating System, Message Interface, and SDK; comprehensive infrastructure configured in a single deployment
- The client AI system invokes perception data via the SDK, and the motion control interface sends all degrees of freedom space positions in one command for the robot to achieve full-body coordinated motion control

Products

FMC³ Brain+ODM



Application Scenarios

- Sorting
- Handling
- Loading & Unloading
- Flexible Grasping

Functions

- Dualarm collaborative mobile manipulation
- Omnidirectional movement + waist lift/pitch
- Multimodal sensing & 1ms fast response
- Quick deployment, outofbox use

Parameters

Configuration		2×7axis arms, 4×vision cameras, 1×mobile chassis, 1×lifting unit
Total DOF	21	
Waist Movement	Lift 0–800 mm; Pitch 0°–90°	
Neck Movement	Rotate ±90°; Pitch 0°–45°	
Arm Payload	5 kg	
Dimensions & Weight	750×600×1565 mm; 200 kg	
Controller	Intel i512th Gen, 128G SSD, WIFI6	
Power & Interface	DC 30–60 V, 1500 W; EtherCAT, 4×GigE LAN	
Environment	IP20, 0–50 °C	
Material	Steel + Aluminum Alloy	



Efficient DualArm Collaboration



DualArm Coordination & Full Workspace Reachability



MultiModal Perception & Safety Assurance

Products

FMC³ Brain+ODM

Industrial
Service Robot



Application Scenarios

- Logistics sorting
- Highprecision assembly & operation
- Industrial inspection
- Guided tours
- Security check
- R&D support

Functions

- Perform highprecision assembly and manipulation
- Move flexibly in all directions
- Interact intelligently with humans
- Execute realtime remote operation
- Ensure safe operation in all scenarios
- Work continuously with autonomous charging
- Support fast application development

Parameters

Body	26 DOF; humanlike motion; 5-DOF waist-leg system
Mobility	4wheel steering; omnidirectional movement; inplace rotation; crab walk; max speed 1.5 m/s
Arms	7DOF arms; $\leq 0.5\text{N}$ force control precision; submillimeter accuracy; 5 kg payload
Hands	Dexterous hand (optional); 3D tactile sensing; quick replacement; full joint torque sensing
Computing	NVIDIA Jetson Thor 5000 (2070 TFLOPS FP4);
Vision	Binocular stereo camera; RGB-D cameras; fisheye array; 360° surround view; 2× LiDAR
Audio	Mic array; sound source localization; 8W speaker
Power	Dual hot-swappable batteries; autonomous charging; 24/7 operation
Safety	360° perception; arm collision detection; active-passive protection
Neck	Roll, Pitch, Yaw
Waist & Legs	3× waist joints; 2× leg joints; 360° coverage

Products

FMC³ Brain+ODM

Humanoid Home
Service Robot



Application Scenarios

- Social & emotional companionship
- Public service & guidance
- Elderly care & assistance
- Rehabilitation & health support
- Family accompanying & interaction
- AI research & development



Functions

- Whole-body teleoperation & human motion mapping
- Precise manipulation with multi-DOF dexterous hands
- Emotional interaction & social companionship
- Assisted care, rehabilitation training support
- 24/7 operation with hot-swap batteries
- Customizable functions



Parameters

Height	165 cm
Weight	71 kg
Total DOF	55
Hand	12-DOF dexterous hand
Arm Payload	3 kg per arm
Actuator	Self-developed highperformance integrated actuator
Battery	Dual hot-swappable
Battery Life	3 hours per battery
Operation	24/7 continuous operation
Vision	RGB + structured-light camera; facial recognition & tracking
Audio	4-mic array; echo cancellation; sound source localization
Tactile	Distributed pressure sensors for touch interaction
Control	Wholebody teleoperation; low-latency motion mapping

Products

FMC³ Brain+ODM

GreetingBot Nova Service Robot



Application Scenarios

- Showrooms & exhibitions
- Reception desks & office buildings
- Museums, galleries, guided tours
- Hotels, hospitals, retail stores

Functions

- Large language model voice interaction in 30+ languages
- Smart reception, intelligent Q&A, guidance and tour explanation
- Dynamic promotion with 14-inch FHD customer-facing screen
- Marker-free deployment, autonomous navigation and obstacle avoidance
- Automatic recharge for continuous front-of-house service

Parameters

Size	558 × 525 × 1350 mm
Net Weight	56 kg
Screen	14-inch; 1080 FHD
Camera	13M high-definition head camera
Mobility	0.5-1.2 m/s adjustable cruising speed
Endurance	12-14 h typical cruising scene
Charging	4.5 h via automatic charging dock or cable
Interfaces	USB 2.0, Type-C, LAN, DC power 12V 3A
Business Value	Reduce routine reception work; improve visitor engagement

Products

FMC³ Brain+ODM

IBen-A03 Service Robot



Application Scenarios

- Banks & government service halls
- Exhibition halls and showrooms
- Reception, greeting and consultation
- Guided tours, entertainment interaction

Functions

- Humanoid design with 10-DOF flexible dual arms
- Reception & greeting with user recognition and custom greetings
- Business consultation via business knowledge base and LLM connection
- Indoor autonomous navigation, leading and real-time obstacle avoidance
- Android SDK / web adaptation for extend development

Parameters

Dimensions	54 × 54 × 155 cm
Weight	79 kg
Arms	5-DOF per arm; shoulder, elbow and finger movement
Screen	15-inch HD touchscreen
Sensors	30 m radar, 3 ultrasonic sensors, 1 depth camera
Compute	RK3588 8-core CPU; 8G+64G
Battery	30Ah LiFePO4; 8 h usage, 6 h charge
Speed	0.3-0.7 m/s adjustable; ±5 cm positioning accuracy

FMC³ Robotics

Thanks

www.fmc3robotics.ai

